Experiment: 5

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Subject Name: DBMS

Aim of the Practical:

To understand and apply conditional logic in SQL by using the modulus operator (MOD / %) to analyze numerical data and classify employee salaries as odd or even, thereby improving data analysis and decision-making skills in SQL.

Tool Used:

Database Management System:
PostgreSQL

Database Administration Tool:
psql (iTerm2 Terminal)

Objective:

To write and execute SQL queries that use the MOD() function to check employee salaries and classify them as odd or even values from an employee table.

Practical / Experimental Steps

Step 1: Open pgAdmin and connect to the PostgreSQL server.

Step 2: Create the Employee table with required attributes.

Step 3: Insert multiple records into the Employee table.

Step 4: Write a SELECT query using CASE statement and MOD() function to classify salaries as odd or even.

Step 5: Execute the query and observe the output.

Step 6: Verify the correctness of the results.

• I / O Analysis

1️⃣ Table Creation

Program:

```
CREATE TABLE Employee(
emp_id INT PRIMARY KEY,
name VARCHAR(50) NOT NULL,
salary INT NOT NULL
);
```

Output:

Table "Employee" created successfully.

2️⃣ INSERT TABLE

Program:

```
dinkygoyal-# (1,  'Amit Sharma',      50000),
dinkygoyal-# (2,  'Priya Verma',      45575),
dinkygoyal-# (3,  'Rahul Mehta',      62000),
dinkygoyal-# (4,  'Sneha Kapoor',     38745),
dinkygoyal-# (5,  'Vikram Singh',     71000),
dinkygoyal-# (6,  'Anjali Desai',     53211),
dinkygoyal-# (7,  'Rohan Gupta',      48000),
dinkygoyal-# (8,  'Neha Reddy',       39999),
dinkygoyal-# (9,  'Karan Malhotra',   84500),
dinkygoyal-# (10, 'Isha Nair',        36781),
dinkygoyal-# (11, 'Arjun Patel',      90000),
dinkygoyal-# (12, 'Meera Joshi',      42133),
dinkygoyal-# (13, 'Siddharth Rao',    76000),
dinkygoyal-# (14, 'Pooja Bansal',     58917),
dinkygoyal-# (15, 'Manish Yadav',     64000);
INSERT 0 15
```

Output:

15 rows inserted successfully into the Employee table.

3️⃣ Display Odd and Even Salaries

Program:

```
SELECT
emp_id AS "Employee ID",
name AS "Employee Name",
salary AS "Salary",
CASE
WHEN MOD(salary,2) = 0 THEN 'Even'
ELSE 'Odd'
END AS "Salary Category"
FROM Employee
ORDER BY salary;
```

Output:

```
dinkygoyal-# ORDER BY salary;
 Employee ID | Employee Name   | Salary | Salary Category
-------------+----------------+--------+-----------------
          10 | Isha Nair       |  36781 | Odd
           4 | Sneha Kapoor    |  38745 | Odd
           8 | Neha Reddy      |  39999 | Odd
          12 | Meera Joshi     |  42133 | Odd
           2 | Priya Verma     |  45575 | Odd
           7 | Rohan Gupta     |  48000 | Even
           1 | Amit Sharma     |  50000 | Even
           6 | Anjali Desai    |  53211 | Odd
          14 | Pooja Bansal    |  58917 | Odd
           3 | Rahul Mehta     |  62000 | Even
          15 | Manish Yadav    |  64000 | Even
           5 | Vikram Singh    |  71000 | Even
          13 | Siddharth Rao   |  76000 | Even
           9 | Karan Malhotra  |  84500 | Even
          11 | Arjun Patel     |  90000 | Even
(15 rows)
```

• Learning Outcomes (What I Have Learnt):

• Understood how to create tables in PostgreSQL.
• Learned how to insert multiple records into a database table.
• Gained knowledge of using the MOD() function to determine odd and even values.
• Learned to apply conditional logic using CASE statements in SQL queries.
• Developed the ability to classify and analyze salary data directly in SQL.
• Improved understanding of SQL-based data analysis and decision-making.